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MPP⁺ (1-methyl-4-phenylpyridine) is a neurotoxin to dopamine-, norepinephrine- and serotonin-containing neurons

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1-Methyl-4-phenylpyridine (MPP⁺) is an oxidative metabolite of 1-methyl-4-phenyl-1,2,3,6-tetrahydroxypyridine (MPTP). MPP⁺ produced local cell death when injected directly into substantia nigra compacta, locus coeruleus, or dorsal and median raphe nuclei of rats. Corresponding significant decreases in dopamine, serotonin and norepinephrine levels were observed in the terminal areas. These observations indicate that MPP⁺ is a non-selective neurotoxin which causes lesions not only in dopaminergic neurons but also in noradrenergic and serotonergic systems following intracranial administration. Selective lesioning of these monoaminergic systems could only be achieved by a stereotaxic injection of MPP⁺ into specific brain regions containing monoamine neurons.

MPP⁺; MPTP; Neurotoxin; Substantia nigra; Raphe nucleus; Locus coeruleus; (Degeneration)

1. Introduction

MPP⁺ is an oxidative metabolite of MPTP which causes a parkinsonian syndrome in both man and monkey (Davis et al., 1979; Burns et al., 1983; Langston et al., 1983; Chiueh et al., 1985). This metabolite of MPTP is formed readily by type B monoamine oxidase (MAO) in the brain (Chiba et al., 1984), accumulating not only in dopaminergic neurons such as nigrostriatal and mesolimbic systems, but also in other monoaminergic areas (Markey et al., 1984). The binding of MPP⁺ to dopamine neuromelanin in the sub-

stantia nigra compacta (A9) neurons (Lyden et al., 1983) may play a major role in the expression of the selective neurotoxic effect of MPTP (Burns et al., 1984). The formation and/or retention of MPP⁺ in the brain (Johannesson et al., 1985; Markey et al., 1984; Heikkila et al., 1984; Cohen et al., 1984; Langston et al., 1984) may account for the species difference in sensitivity to MPTP's neurotoxicity (Chiueh et al., 1984a).

Direct infusion of MPP⁺ (Bradbury et al., 1986; Sun et al., 1985), but not of MPTP (Chiueh et al., 1984b) into rat substantia nigra causes a marked reduction of striatal dopamine levels. However it is unclear whether this neurotoxin's effect is specific to dopaminergic neurons or whether all monoaminergic neurons are sensitive to MPP⁺.

In this study, we examined the selectivity of MPP⁺-induced neurotoxicity by injecting it into three different monoaminergic nuclei: substantia nigra compacta, locus coeruleus, and dorsal and medial raphe nuclei of rats.

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2. Materials and methods

Male Sprague-Dawley rats (280-320 g, Zivic-Miller Farms, Allison Park, PA) were anesthetized by intraperitoneal (i.p.) injection of chloral hydrate (40 mg/kg; Sigma, St. Louis, MO). Eight micrograms of MPP⁺ (4 µg/µl saline; RBI, Wayland, MA) were injected stereotactically (David Kopf Instrument, Palo Alto, CA; Paxinos and Watson, 1982) at the rate of 1 µl/5 min through a 30 gauge needle placed into either the left substantia nigra (AP: 4.0 mm, Lat: 2.0 mm, DV: 2.0 mm from interaural zero line) or the left nucleus locus coeruleus (AP: -0.5 mm, Lat: 1.1 mm, DV 2.5 mm). For the injection of MPP⁺ into the raphe nucleus, 4 µg of MPP⁺ was injected into the dorsal raphe (AP: 1.2 mm, Lat: 0 mm, DV: 3.3 mm) and 4 µg was injected into the median raphe (AP: 1.2 mm, Lat: 0 mm, DV: 1.3 mm). Sham controls consisted of rats in which saline was injected directly into the substantia nigra, locus coeruleus or raphe nuclei. In some rats the MPP⁺ dosage was decreased from 8 µg to 4 or 2 µg to determine the dose-dependency of MPP⁺'s effect on the nigrostriatal dopaminergic neurons.

After recovery from anesthesia, animals were housed in plastic cages (4 per cage; 60 × 30 × 30 cm³) and observed daily up to 14 days for behavioral changes, e.g. rotational behavior (Ungerstedt, 1973) and serotonin syndrome (Jacobs, 1976). Spontaneous and d-amphetamine-induced (3 mg/kg i.p., 35-40 min; Sigma, St. Louis, MO) rotational behavior was measured in a square cage (30 × 30 × 30 cm³) by counting the number of 360° complete rotations around the cage in the direction of the lesion side (ipsilateral) and/or the intact side (contralateral) in 5 min intervals on 1, 2, 7 and 10 days following intranigral injection of MPP⁺. No attempt was made to quantify the serotonin syndrome which was only observed on the first day following intraraphe MPP⁺ injection.

After 14 days, rats were killed by decapitation and the brains were microdissected. Tissue samples from the striatum, cingulate cortex, olfactory tubercle and nucleus accumbens were assayed for monoamines by using a high performance liquid chromatographic method (HPLC) with electrochemical detection as previously described (Chiueh

et al., 1983; 1984b). The frozen midbrain and medulla oblongata of each animal were cut into 20 µm thick sections and were mounted on gel-coated slides. Serial sections through substantia nigra, locus coeruleus or raphe nuclei were processed for thionin staining to determine the extent of degeneration.

Results are presented as mean ± one S.E.M. of at least 6 animals. Results were analyzed by Student's t-test. A probability (P) value of less than 0.05 was considered statistically significant.

3. Results

After recovery from anesthesia, rats with an intranigral MPP⁺ (8 µg) injection showed rotation away from the MPP⁺-injected side. These spontaneous rotation scores were 9.4 ± 1.7 , 11.9 ± 3.9 and 5.2 ± 3.9 rotations in 5 min, respectively when measured 1, 2, 7 and 10 days after injection. Ten days after injection, some rats with nigral MPP⁺ injection (8 µg) were given a 3 mg/kg i.p. administration of d-amphetamine. All of these d-amphetamine-treated rats showed significant ipsilateral rotations towards the lesion side (80 ± 15 rotations in 5 min, $P < 0.05$).

Rats with raphe MPP⁺ injection showed straub tail, tremors, and retropulsion limited to the first day after recovery from anesthesia. No motor asymmetry nor serotonin-like syndrome was noticed in the unilateral locus coeruleus lesion group.

A dose-dependent decrease in striatal dopamine was observed following intranigral administration of MPP⁺. Dopamine levels decreased by about 30, 70 and 99%, respectively in the ipsilateral striatum after injection of 2, 4 or 8 µg MPP⁺ to the substantia nigra. Intranigral MPP⁺ also reduced serotonin and norepinephrine levels in the cingulate cortex (table 1). This observation indicates that ascending serotonin and norepinephrine fibers were also damaged by MPP⁺ because these fibers pass dorsally to the substantia nigra, and then merge with ascending dopamine fibers to form the medial forebrain bundle (Ungerstedt, 1973). This MPP⁺-induced cytotoxic damage is confirmed by the histological examination of the brain slices (fig. 1A), in which the

TABLE 1

Monoamine levels in the striatum and cingulate after a unilateral injection of MPP⁺ into the substantia nigra compacta. Male Sprague-Dawley rats (280-320 g, N = 6) were killed 14 days after MPP⁺ injection (8 µg in 2 µl) into the left substantia nigra compacta. Monoamine levels (serotonin = 5HT; norepinephrine = NE; dopamine = DA) in micropunched cingulate cortex and striatum of both sides (left = lesion side) were measured by HPLC using electrochemical detection. Values (pg/µg protein) are expressed as the mean ± one S.E. The % decrease column shows decreased percentage of mean value in the lesioned side as compared with the control. P: Student's paired t-test.

Monoamine (pg/µg protein)	Area	Control side	Lesion side	Significance	Decrease (%)
5HT	Striatum	3.13 ± 0.12	1.11 ± 0.10	P < 0.01	65
	Cingulate	2.07 ± 0.15	0.49 ± 0.01	P < 0.01	76
NE	Cingulate	3.77 ± 0.26	2.19 ± 0.50	P < 0.05	41
DA	Striatum	160 ± 15.65	1.20 ± 0.10	P < 0.01	99
	Cingulate	2.20 ± 0.17	0.75 ± 0.09	P < 0.01	66

TABLE 2

Monoamine levels in the striatum, nucleus accumbens and cingulate cortex following a unilateral injection of MPP⁺ into the locus coeruleus. Rats were killed 14 days after MPP⁺ injection to the left locus coeruleus. Monoamine levels (serotonin = 5HT; norepinephrine = NE; dopamine = DA) in micropunched brain tissues of control and lesioned sides were measured by a HPLC procedure. Values (mean ± S.E.) were expressed by pg/µg protein. P: Student's paired t-test. N = 6.

Monoamine (pg/µg protein)	Area	Control side	Lesion side	Significance	Decrease (%)
5HT	Striatum	2.76 ± 0.25	2.83 ± 0.29	NS	—
	Cingulate	2.85 ± 0.47	3.07 ± 0.32	NS	—
NE	Cingulate	3.13 ± 0.19	0.99 ± 0.05	P < 0.01	68
	Accumbens	27.0 ± 3.32	13.1 ± 1.70	P < 0.01	51
DA	Striatum	165 ± 12.34	153 ± 13.95	NS	—

TABLE 3

Monoamine levels in the striatum, olfactory tubercle (Olf tub) and cingulate cortex after injection of MPP⁺ into raphe nucleus. MPP⁺ (4 µg in 1 µl) was injected to both dorsal and median raphe nuclei (lesion group). Saline was injected into raphe nucleus of animals in the sham control group. Rats were killed 14 days after injection. Monoamine levels (serotonin = 5HT; norepinephrine = NE; dopamine = DA) in micropunched brain tissues were measured by a HPLC procedure. Values (mean ± S.E.) are expressed as pg/µg protein. N = 6. P: Student's grouped t-test.

Monoamine (pg/µg protein)	Area	Control group	Lesion group	Significance	Decrease (%)
5HT	Striatum	2.70 ± 0.28	0.88 ± 0.19	P < 0.01	67
	Cingulate	2.83 ± 0.35	0.59 ± 0.22	P < 0.01	79
	Olf tub	12.8 ± 1.44	3.80 ± 0.41	P < 0.01	70
NE	Cingulate	3.39 ± 0.27	3.03 ± 0.10	NS	—
	Olf tub	4.09 ± 0.21	3.49 ± 0.57	NS	—
DA	Striatum	184 ± 5.77	163 ± 18.80	NS	—
	Cingulate	1.90 ± 0.24	1.86 ± 0.22	NS	—

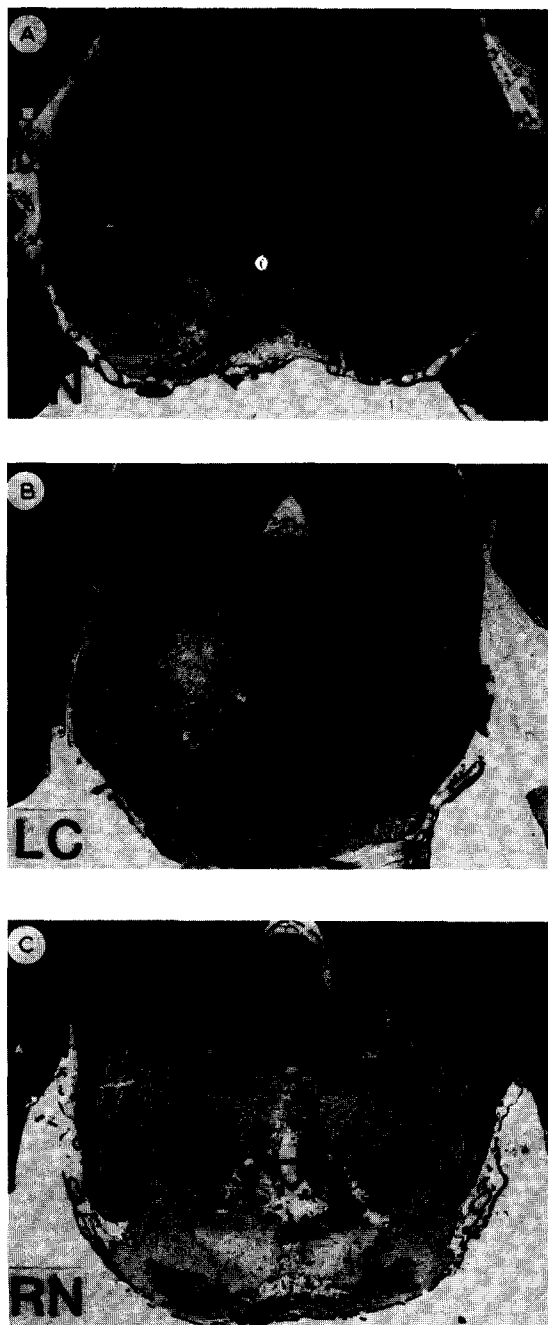


Fig. 1. Intracranial administration of MPP^+ causes cytotoxic damage in: (A) the substantia nigra compacta (SN); (B) the locus coeruleus (LC); and (C) the dorsal and median raphe nuclei (RN). The frozen midbrain and medulla oblongata were cut into $20\ \mu\text{m}$ thick sections. Brain slices were processed for thionin staining. Photos are enlarged by 14X. Arrows depict MPP^+ -induced lesions.

whole substantia nigra area is necrotized, following the intranigral administration of MPP^+ .

Locus coeruleus lesions by MPP^+ produced only 50-70% loss of norepinephrine in the cingulate cortex and in the nucleus accumbens (table 2). Serotonin and dopamine in both areas on the lesioned side were preserved. This lack of effect of locus coeruleus lesion on the dopamine and serotonin content of cingulate cortex and nucleus accumbens can be attributed to the absence of ascending dopaminergic or serotonergic fibers in the vicinity of the locus coeruleus.

The cingulate cortex and olfactory tubercle receive serotonin innervation from the median raphe nucleus, whereas the striatum receives its innervation from the dorsal raphe nucleus (Parent et al., 1981). Serotonin levels were decreased by 70-80% in all of these terminal areas following the MPP^+ infusion into both dorsal and median raphe nuclei (table 3). It would appear to be difficult to lesion all of the serotonergic cell bodies since they are distributed widely throughout the brain-stem (Singhaniyom, 1984). In the raphe-lesioned rats, norepinephrine and dopamine levels were not different from the control, which probably reflects the fact that these ascending catecholamine bundles do not pass through the raphe nucleus.

Histological examination revealed that MPP^+ caused cytotoxic damage to the brain tissue only at the injection sites, i.e. substantia nigra, locus coeruleus and raphe nuclei (fig. 1).

4. Discussion

A direct application of MPTP to non-pigmented substantia nigra dopaminergic neurons of rats produces no significant change in striatal dopamine (Chiueh et al., 1984b). In the current study, intranigral administration of MPP^+ (2-8 μg) caused a dose-dependent decrease of dopamine in its major terminal area, the striatum. The application of MPP^+ to the ventro tegmental A10 dopamine neurons also decreased the dopamine content in the mesolimbic projection areas (Namura, unpublished observation). Thus, MPP^+ 's neurotoxic effect is not specific to nigrostriatal dopaminergic neurons as previously suggested (Markey et al., 1984; Bradbury et al.,

1986; Irwin and Langston, 1984). Furthermore, intracranial administration of MPP^+ to the major raphe nuclei or locus coeruleus also caused damage to the ascending serotonergic system and the noradrenergic system, respectively. Thus, MPP^+ lesions not only dopaminergic neurons, but also serotonergic and noradrenergic neurons. Specific or 'selective' lesioning of these monoaminergic systems was only achieved by injection of MPP^+ into selective anatomical areas, such as a specific cell body group or compact fiber bundles.

It is possible that the effect of MPP^+ on monoaminergic neurons is non-specific at these doses (2-8 μ g) as MPP^+ may act as a cationic surfactant to dissolve the cell membranes. Necrotic lesion was observed histologically (fig. 1) at the site of injection which resembles the lesion produced by the procedure of electrical coagulation and/or by the intracerebral injection of strong acids or bases. Other substances containing pyridinium ion are used as cationic surfactants or antiseptics (see review of Markey and Schmitt, 1986).

Systemic administration of MPP^+ to rodents produces no significant changes in brain dopamine levels but causes sympathetic discharge, cataract and lung lesions resembling the effects of Paraquat (Johannessen et al., 1986). In fact, MPP^+ has been tested as a herbicide under the trade name Cyperquat (Gulf Oil Chemical Company, Merriam, KA). These effects may be due to the conversion of pyridinium ion to the free radical causing oxygen stress and/or damage in certain tissues (see review of Markey and Schmitt, 1986).

In the current study, no apparent behavioral changes or imbalances were observed in the unilateral locus coeruleus lesion group. However, spontaneous contralateral turning behavior, lasting more than 7 days, was observed immediately following a unilateral intranigral injection of MPP^+ . When d-amphetamine was administered to these nigral MPP^+ -lesioned animals at 10 days, the direction of rotation was changed to the lesioned (ipsilateral) side. This d-amphetamine-induced circling may be the result of increased dopamine release from the nigrostriatal neuron of the intact side as compared to the lesioned side. Similar rotational behavior in rats has been reported following administration of 6-hydroxy-

dopamine into the substantia nigra (Ungerstedt, 1973). These behavioral results indicate that unilateral MPP^+ -induced lesioning of the nigrostriatal pathway was almost complete at this time.

The acute spontaneous contralateral turning behavior of rats observed following a unilateral administration of MPP^+ into the substantia nigra area may result from an acute increase in the release of nigrostriatal dopamine from the injected side. It has been shown that systemic administration of MPTP increases the release of plasma catecholamines (Chiueh et al., 1984a). In the brain synaptosomal preparation, both MPTP and MPP^+ also increase the release of endogenous dopamine (Chiueh et al., unpublished observation). These effects may be due, partially, to competition for the uptake of dopamine (Denton and Howard, 1984; Javitch et al., 1985). In addition, some release of neurotransmitters may be due partially to a leakage of the contents of synaptic vesicles during the degeneration period after MPP^+ treatment.

A previous study has shown that systemic administration of MPTP causes an acute serotonin syndrome due to a rapid increase in brain serotonin levels (Chiueh et al., 1984b). Retropulsion, tremor and straub tail in rats were also demonstrated in the current experiment for at least 24 h following intrapaphe administration of MPP^+ . The observation that either systemic MPTP or intrapaphe MPP^+ causes serotonin activation seems reasonable since serotonergic neurons contain mainly type B MAO (Westlund et al., 1985) which is responsible for the conversion of MPTP to MPP^+ (Chiba et al., 1984). In the past, this serotonin syndrome could only be induced following combination of MAO inhibition with tryptamine or 5-hydroxytryptophan treatment (Jacobs, 1976; Squires, 1978).

In conclusion, the current results do not substantiate the assertion that MPP^+ is a selective or specific nigrostriatal dopamine neurotoxin. The mechanism of MPTP's selective neurotoxicity on the nigrostriatal dopaminergic neuron remains to be solved. Nevertheless, a relatively selective lesion of the specific monoaminergic cell bodies could be achieved by direct intracranial administration of MPP^+ .

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